

Shamba Voices

Voice-First AI Agronomy Co-Pilot & Green Biotech Solution for East Africa
Wavumbuzi Challenge Submission Spec

1. Executive Overview & Problem Context

Over 60% of East Africa's population relies on smallholder farming. However, farmers lose 30% to 40% of their crop yields annually due to delayed pest detection and lack of accessible diagnostic tools. Existing solutions require high-speed smartphones, English typing skills, or downloading heavy mobile applications. **Shamba Voices** bridges this accessibility gap by introducing a zero-download, multilingual **Voice-In / Voice-Out AI Engine** built on top of WhatsApp and USSD/SMS gateways.

2. Technical Architecture & Component Requirements

The system architecture consists of four modular layers designed for high speed and near-zero marginal operating cost:

Layer	Technology Stack	Core Functionality
Ingestion / Channel Gateway	Twilio WhatsApp API / Africa's Talking USSD Gateway	Receives incoming audio notes (.ogg/.mp3), images (.jpeg), and text messages from farmers without requiring an app installation.

Layer	Technology Stack	Core Functionality
Voice Processing Engine	OpenAI Whisper API / Speech-to-Text	Transcribes spoken regional dialects (Swahili, Kikuyu, Luo, Kinyarwanda) into clean structured text prompts.
Diagnostic & Biotech AI Model	Vision LLM + Fine-tuned Plant Pathogen Prompt Logic	Analyzes plant leaf images and text queries to identify pests (e.g., Fall Armyworm) and prescribe organic bio-pesticides.
Voice Output Generator	ElevenLabs / Google Cloud Text-to-Speech (Swahili)	Converts diagnostic and market price text answers into natural spoken Swahili voice notes delivered back to WhatsApp.

3. Product Requirements & Features

- Zero-App Onboarding:** Farmers interact directly via WhatsApp or basic USSD feature phone menus.
- Multilingual Voice Processing:** Native support for Swahili audio queries and replies to eliminate literacy barriers.
- Green Biotechnology Prescriptions:** Recommends biological control remedies (e.g., *Bacillus thuringiensis*, Neem extracts, copper-based fungicides) to restore soil health and prevent pest resistance.
- Predictive Market Price Analytics:** Real-time price notifications connecting smallholder farmers with buyers in regional aggregation centers.

4. Core Logic Code Snippet (Python Prototype)

Below is the core diagnostic and speech processing workflow script for Shamba Voices:

```
import os
import requests

def process_shamba_voices_query(user_audio_url, user_image_url=None):
    """
    Shamba Voices Processing Pipeline:
    1. Transcribe Swahili Audio to Text
    2. Analyze Crop Pathogen / Pest Issue
    3. Generate Green Biotech Remedy
    4. Convert Reply to Swahili Audio
    """
    # Step 1: Transcribe incoming Swahili voice note
    transcribed_text = transcribe_swahili_audio(user_audio_url)

    # Step 2: System Prompt for Diagnostic AI
    system_prompt = """
    You are Shamba Voices, a voice-first agricultural AI assistant for East
    Africa.
    - Diagnostic Focus: Identify crop diseases and pests.
    - Solution Rules: Always prioritize organic, biological pest controls
    (e.g., Neem, Bt, cover crops).
    - Language: Respond in simple, encouraging Swahili suitable for voice
    playback.
    """

    # Step 3: Run AI Model Diagnosis
    diagnosis_response = run_ai_diagnostics(system_prompt, transcribed_text,
user_image_url)

    # Step 4: Synthesize Swahili Voice Reply
    voice_note_url = generate_swahili_audio_reply(diagnosis_response)

    return {
        "text_response": diagnosis_response,
        "audio_response_url": voice_note_url
    }
```

5. Deployment & GitHub Pages Implementation

To publish the frontend simulator for your Wavumbuzi Challenge submission, follow these deployment steps:

1. Save your web code as index.html.
2. Create a public repository named shamba-voices on GitHub.
3. Upload index.html to the root repository folder.
4. Navigate to **Settings > Pages**, set the source branch to main, and click **Save**.
5. Access your live platform link at <https://<your-username>.github.io/shamba-voices/>.

Shamba Voices - Wavumbuzi Venture Concept

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